

Electric Charges & Fields

Properties of Electric Charge

a) Additivity of Charge: The total charge of a system is the algebraic sum of individual charges present in system.

If it contains multiple charges $q_1, q_2, q_3 \dots$
 $Q = q_1 + q_2 + q_3 \dots$

Example: If a body has, $+5C, -3C, +2C$
 then it will be $Q = 5 + (-3) + 2$
 $= 4C$.

b) Quantisation of Charge: When physical quantity can assume only discrete values

Charge $e (= 1.6 \times 10^{-19} C)$

Where Charge is given by $q = ne$
 but 'n' can not be fractional

c) Conservation of Charge: The total Charge in an isolated system remains constant.
 It can neither be created nor destroyed.
 but it can be transferred from one body to another.

$\Rightarrow$ When a Glass rod is rubbed with silk electrons are transferred from glass body to the silk.

COULOMB'S LAW (Force between two point charges)

$\Rightarrow$ two stationary point charges repel or attract each other with a force which is directly proportional to the product of charges and inversely proportional to the square of the distance between them.

$$F \propto \frac{q_1 q_2}{r^2}$$

$$F = k \frac{q_1 q_2}{r^2} \quad [k \text{ in air} = 9 \times 10^9]$$

$$\left[k = \frac{1}{4\pi\epsilon_0} \right] \Rightarrow F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$$

$\Rightarrow$ Permittivity of free space (ϵ_0)

$$\epsilon_0 = 8.854 \times 10^{-12} \text{ F m}^{-1}$$

$\Rightarrow$ Absolute Permittivity (ϵ)

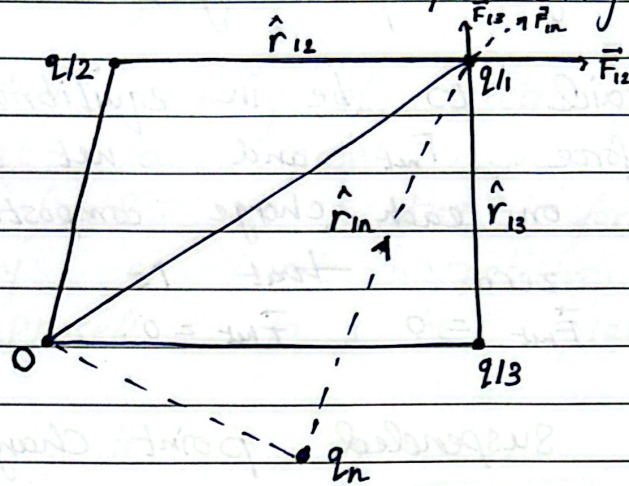
It measures its ability to allow the establishment of an electric field within it.

$\Rightarrow F \rightarrow \text{attractive} \quad q_1 q_2 < 0$
 $F \rightarrow \text{repulsive} \quad q_1 q_2 > 0$

$\Rightarrow$ Dielectric constant (Relative Permittivity)

$$= K = \frac{\epsilon}{\epsilon_0} = \frac{F_{\text{vacuum}}}{F_{\text{medium}}}$$

* Forces between multiple charges : Superposition Principle



If a system contains a number of interacting charges, then the net force on any one charge due to the other charges is the 'vector' sum of all the forces exerted on it by all the other charges. The individual forces are unaffected due to presence of other charges.

$$\vec{F}_1 = \vec{F}_{12} + \vec{F}_{13} + \dots + \vec{F}_{1n}$$

$$\Rightarrow \vec{F}_{12} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{12}^2} \hat{r}_{12}$$

$$\Rightarrow \vec{F}_{13} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_3}{r_{13}^2} \hat{r}_{13}$$

$$\vec{F}_1 = \frac{1}{4\pi\epsilon_0} \left[\frac{q_1 q_2}{r_{12}^2} \hat{r}_{12} + \frac{q_1 q_3}{r_{13}^2} \hat{r}_{13} + \dots + \frac{q_1 q_n}{r_{1n}^2} \hat{r}_{1n} \right]$$

* Equilibrium of a system of Charges

i) A system is said to be in equilibrium if the net force F_{net} and net torque τ_{net} acting on each charge composing the system is zero that is

$$\vec{F}_{net} = 0 \quad \vec{\tau}_{net} = 0.$$

ii) Equilibrium of suspended point charge:
 Two balls having charge $+q$ by string length 'l'.

$$\boxed{\tan \theta = \frac{F_e}{mg}}$$

iii) $\Rightarrow$ An equilibrium is considered stable if a slight displacement results in bringing system back to original state.

$\Rightarrow$ If it deviates further from equilibrium it is unstable

$\Rightarrow$ If it produces no displacement it is neutral.

* Concept of Electric Field.

$\Rightarrow$ The space surrounding an electric charge q in which another charge q_0 experiences a electrostatic force of attraction or repulsion is called electric field charge q .

* Intensity of Electric Field.

The intensity of electric field due to a charge at a point in space may be defined as the force experienced by a unit positive point charge supposed to be placed at that point.

It is given by

$$\vec{E} = \frac{\vec{F}}{q_0}$$

* Electric Field Intensity due to a Point Charge.

It is given by :

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \text{ N/C}$$

$$\Rightarrow E = k \frac{q}{r^2}$$

$$\Rightarrow \vec{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r} \quad (\text{in vector form})$$

$$\text{(Force)} \quad F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} = \frac{k q^2}{r^2}$$

$$\text{(Potential Energy)} \quad U = \frac{1}{4\pi\epsilon_0} \frac{q^2}{r} = \frac{k q^2}{r}$$

$$\text{(Potential/Voltage)} \quad V = \frac{1}{4\pi\epsilon_0} \frac{q}{r} = \frac{k q}{r}$$

★ Intensity of Electric Field due to a Continuous charge Distribution.

i) Linear Charge Distribution:

$$\text{Linear Charge Density } (\lambda) = \frac{dq}{dl}$$

$$df = \frac{k q_0 dq}{r^2}$$

$$df = \frac{k q_0 \lambda dl}{r^2}$$

$$f = k q_0 \int \frac{\lambda dl}{r^2}$$

$$\boxed{E = k \int \frac{\lambda dl}{r^2}}$$

ii) Surface Charge Distribution:

$$\text{Surface Charge Density } (\sigma) = \frac{dq}{dA}$$

$$df = \frac{k q_0 dq}{r^2}$$

$$df = \frac{k q_0 \sigma dA}{r^2}$$

$$f = k q_0 \int \frac{\sigma dA}{r^2}$$

$$\boxed{E = k \int \frac{\sigma dA}{r^2}}$$

iii] Volume Charge distribution :

$$\text{Volume Charge Density } (\rho) = \frac{dq}{dv}$$

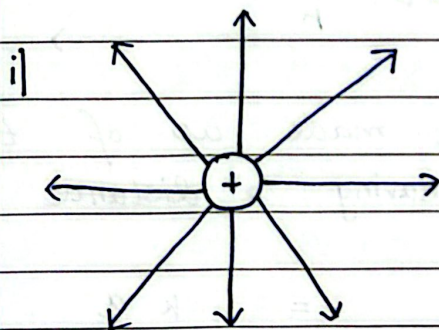
$$df = \frac{kq_0 dq}{r^2}$$

$$df = \frac{kq_0 \rho dv}{r^2}$$

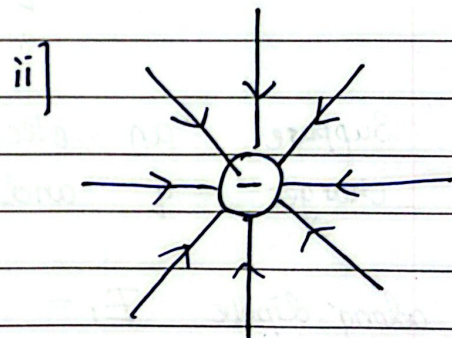
$$f = kq_0 \int \frac{\rho dv}{r^2}$$

$$E = k \int \frac{\rho dv}{r^2}$$

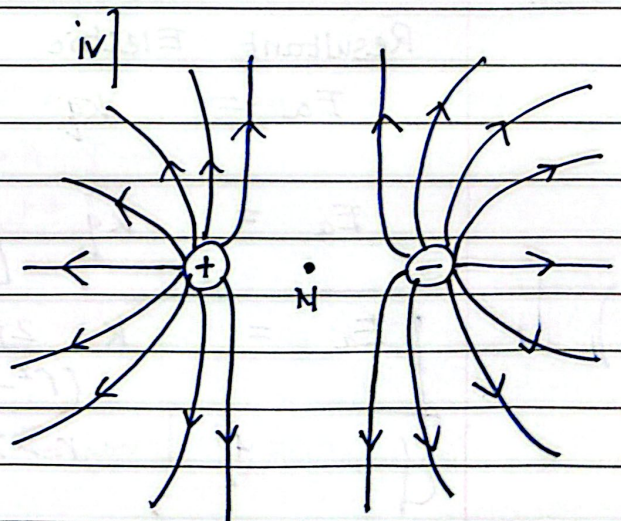
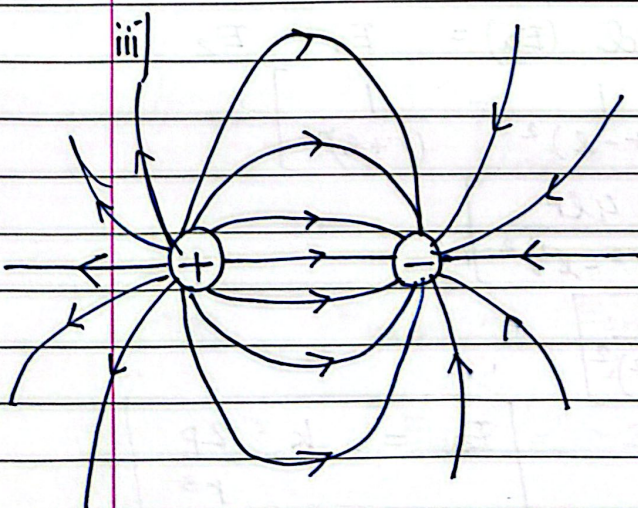
★ Electric Lines of Force.



(+ve q charge)

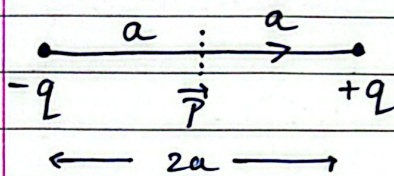


(-ve q charge)



* Electric Dipole

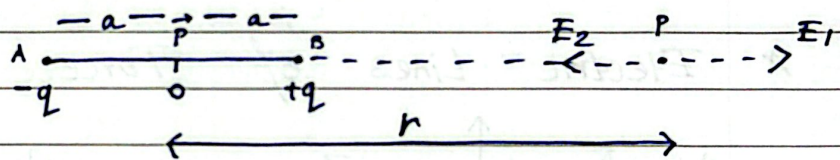
A pair of equal and opposite point charges separated by a vector distance is called an electric dipole



$$\vec{p} = q \times 2a$$

*** Electric Field due to an Electric Dipole.

i) At axis of a Dipole:



Suppose an electric dipole made up of two charges $-q$ and $+q$, having a distance $2a$.

along dipole $E_1 = \frac{1}{4\pi\epsilon_0} \frac{q}{(r-l)^2} = \frac{kq}{(r-l)^2}$

opposite dipole $E_2 = \frac{1}{4\pi\epsilon_0} \frac{q}{(r+l)^2} = \frac{kq}{(r+l)^2}$

Resultant Electric field (E_a) = $E_1 - E_2$

$$E_a = kq \left[\frac{1}{(r-l)^2} - \frac{1}{(r+l)^2} \right]$$

$$E_a = kq \left[\frac{4lr}{(r^2-l^2)^2} \right]$$

$$E_a = k \frac{2pr}{(r^2-l^2)^2}$$



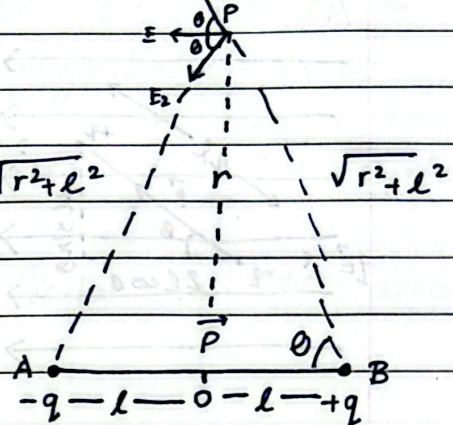
if

$$r \gg l$$

$$E_a = k \frac{2p}{r^3}$$

* ii] At a point on Equatorial line of Dipole

Suppose that point P is situated on right-bisector of the dipole AB at a distance 'r' metre.



$$E_1 = \frac{1}{4\pi\epsilon_0} \frac{q}{(r^2 + l^2)} = \frac{kq}{(r^2 + l^2)} \text{ along } \overrightarrow{BP}$$

$$E_2 = \frac{1}{4\pi\epsilon_0} \frac{q}{(r^2 + l^2)} = \frac{kq}{(r^2 + l^2)} \text{ along } \overrightarrow{PA}$$

$$E = E_1 \cos \theta + E_2 \cos \theta.$$

$$= \frac{kq}{(r^2 + l^2)} \cos \theta + \frac{kq}{(r^2 + l^2)} \cos \theta.$$

$$= (E_1 + E_2) \cos \theta.$$

$$= \frac{2kq}{\sqrt{r^2 + l^2}} \times \frac{l^2}{\sqrt{r^2 + l^2}}$$

$$E_{\text{net}} = \frac{pk}{(x^2 + a^2)^{3/2}}$$

or

$$E_{\text{net}} = \frac{pk}{(r^2 + l^2)^{3/2}}$$

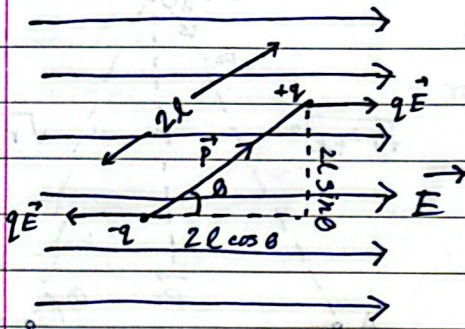
If $r \gg l$ the E_{net} is given by

$$E_{\text{net}} = \frac{pk}{r^3}$$

$$\text{also, } E_{\text{equator}} = \frac{E_{\text{axial}}}{2}$$

$$\Rightarrow E_{\text{axial}} = 2 \times E_{\text{equator}}$$

→ Motion of An Electric Uniform Dipole in Uniform Field.



consider an Electric Dipole Placed in uniform $\vec{E}$ electric field.

$\tau = \text{force} \times \text{perpendicular distance}$

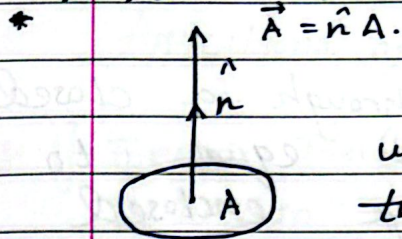
$$\tau = q \cdot E (2l \sin \theta)$$

$$\tau = 2 q l E \sin \theta.$$

$$\tau = p \cdot E \sin \theta$$

$$\vec{\tau} = \vec{p} \times \vec{E}$$

⇒ Area Vector.



Area Vector is $\vec{A} = A \cdot \hat{n}$
 where A is magnitude of
 the area and $\hat{n}$ is the
 unit normal vector to the surface.

Area vector is always perpendicular to the area surface.

$$\Phi_E = \oint \vec{E} \cdot d\vec{A}$$

The electric flux is a measure of number of lines of force passing through some surface held in the electric field.

$$\Phi_E = E A \cos \theta$$

Special Cases. : i] If $\theta = 0^\circ$

$$\Phi_E = E A \cos 0^\circ = E A$$

→

ii] If $\theta = 90^\circ$

$$\Phi_E = E A \cos 90^\circ = 0$$

→

If $\theta = 180^\circ$

$$\Phi_E = E A \cos 180^\circ = -E A$$

Electric Flux Density : is ratio of Electric flux through a surface area

$$= \frac{\Phi_E}{A}$$

Dimensions of Electric Flux : $[ML^3 T^{-3} A^{-1}]$

⇒ * GAUSS THEOREM

*

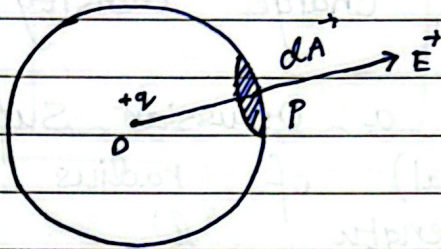
The total electric flux through a closed surface (Gaussian surface) is equal to $\frac{1}{\epsilon_0}$ times the total charge enclosed within the surface.

⇒ Gaussian Surface and Its Properties.

- i) It should be a Closed Surface
- ii) It should pass through the point where electric field is to be calculated.
- iii) In a Gaussian surface if $\text{flux} = 0$, does not imply $E = 0$, but if $E = 0$ flux will be 0.
- iv] Electric Flux depends only inside the charges present in it.

Applications of Gauss' Theorem

- * i] Electric Field due to a Point Charge:
 Deduction of Coulomb's Law from Gauss Theorem



$$\Phi_E = \oint E \cdot dA \cos 0^\circ$$

$$\Phi_E = \oint E \cdot dA$$

$$\Phi_E = E \oint dA$$

$$\Phi_E = E \cdot 4\pi r^2$$

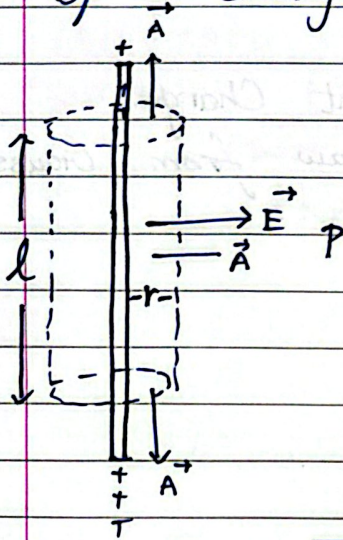
$$\frac{q}{\epsilon_0} = E \cdot 4\pi r^2$$

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$

$$F = E \cdot q_0$$

$$F = \frac{1}{4\pi\epsilon_0} \times \frac{qq_0}{r^2}$$

* ii] Electric Field due to an infinite line of Charge:



Consider an infinitely long line of charge having linear charge density (λ)

consider a Gaussian Surface (cylindrical) of radius 'r' and length 'l'.

To find: Electric Field at P

For Plane Surfaces : $\vec{E} \perp \vec{A}$, $\Phi_E = 0$.

For Curved Surfaces : $\Phi_E = \int \vec{E} \cdot d\vec{A} \cos \theta$
 $= \int E \cdot dA$
 $= E \int dA$

(using Gauss' Theorem)

$$\Phi_E = E \cdot 2\pi r l$$

$$\frac{q}{\epsilon_0} = E \cdot 2\pi r l$$

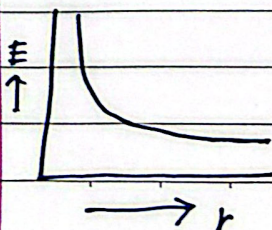
$$q = E \cdot 2\pi r l \epsilon_0$$

$$[q/l = \lambda]$$

$$\boxed{\frac{\lambda}{\epsilon_0 2\pi r} = E}$$

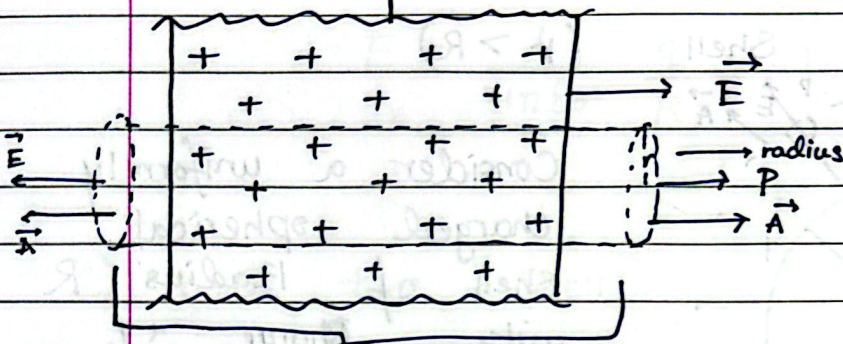
Thus, $\vec{E}$ is inversely proportional to radius

Relation between E and r



$\Rightarrow$ Graph.

* iii] To find Electric Field due to an infinite $\vec{A}$ Plane sheet of Charge.



Consider an infinite plane sheet of charge with surface charge density $-\sigma$

l. Consider a Cylindrical Gaussian

To find E, F at point P .

For Curved Surface : $\vec{E} \perp d\vec{A}$ $\Phi_E = 0$.

For Plane Surface : $2 \times \Phi_E$

$$\Phi_E = 2 \times \oint \vec{E} \cdot d\vec{A} \cos 0^\circ$$

$$\Phi_E = 2 \times E \oint dA$$

$$\Phi_E = 2 \times E \times A$$

Using Gauss' Theorem, $\Phi_E = \frac{q}{\epsilon_0}$

$$2 \times E \times A = \frac{q}{\epsilon_0}$$

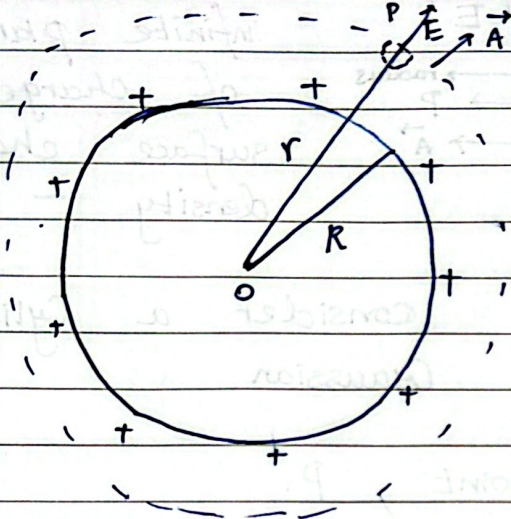
$$[\sigma = q / l^2 \text{ or Area}]$$

$$E = \frac{q}{2A\epsilon_0}$$

$$E = \frac{\sigma}{2\epsilon_0}$$

* iv] Electric Field at a Point due to thin uniformly charged spherical Shell

— outside the Shell ($r > R$)



Consider a uniformly charged spherical Shell of Radius R with charge $+q$, uniformly distributed on its surface.

Consider a Gaussian (concentric spherical) surface of radius through P

Electric Flux linked with Gaussian Surface

$$\Phi_E = \oint \vec{E} \cdot d\vec{A} \cos 0^\circ$$

$$\Phi_E = E \oint dA$$

$$\Phi_E = E \cdot 4\pi r^2$$

Using Gauss' Theorem

$$\Phi_E = \frac{q}{\epsilon_0}$$

$$E \cdot 4\pi r^2 = \frac{q}{\epsilon_0}$$

$$E = \frac{q}{4\pi\epsilon_0 r^2} = \frac{\sigma 4\pi R^2}{4\pi\epsilon_0 r^2}$$

$$= \frac{\sigma R^2}{\epsilon_0 r^2}$$

* b) on the Shell ($r = R$)

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{R^2} \quad E = \frac{\sigma R^2}{\epsilon_0 R^2} = \boxed{\frac{\sigma}{\epsilon_0}}$$

c) Inside the Shell

Consider a point P inside the Shell.
 Draw a Gaussian surface passing through P.
 since this surface doesn't have any charge - Electric field at every point inside the shell is zero.

Relation between E and R.

